

Yasirah Mohamed Ilyas

SUMMARY

Chemical Engineering graduate with ongoing industrial experience through an internship at the Borouge Innovation Centre and a strong foundation in process engineering, process simulation, material and energy balances, and HAZOP. Passionate about applying engineering principles to solve real-world process challenges while continuing to develop my technical expertise in process engineering, process optimization, production, and industrial operations.

EDUCATION

- **Bachelor of Science in Chemical Engineering | Khalifa University, Abu Dhabi, UAE | Graduated: 2026**
- **International School of Choueifat, Ruwais, UAE | Graduated: 2022**

RELEVANT EXPERIENCE

Engineering Intern

Borouge Innovation Centre | Abu Dhabi, UAE | June 2026 – Present

- Currently undertaking a 16-week internship with exposure to polymer research, laboratory workflows, and product development activities.
- Gaining understanding of polymer characterization through verbal instruction and observation, including rheology, thermal analysis, microscopy, and density measurement.
- Learning how polymer properties influence material selection, product performance, and industrial applications.
- Developing awareness of laboratory quality systems, HSE practices, and engineering safety procedures used in petrochemical R&D environments.
- Introduced to procurement processes, material management, inventory systems, and supporting functions such as warehouse logistics and SAP-based tracking.
- Gaining foundational understanding of intellectual property, patents, and innovation protection in polymer development.
- Learning about engineering safety systems including fire alarms, HVAC panels, chiller systems, and emergency response protocols.

Senior Design Project I & II – Lithium Recovery from Waste Batteries using emerging technologies | Khalifa University | 2025 – 2026

- Designed an integrated lithium recovery process using pretreatment, capacitive deionization (CDI), forward osmosis (FO), and post-treatment stages to produce lithium carbonate.
 - Developed a complete Process Flow Diagram (PFD) and process configuration for a scalable lithium recovery plant.
 - Evaluated alternative process equipment using technical decision matrices to select suitable technologies based on performance, feasibility, and sustainability criteria.
 - Performed material and energy balances to validate process performance and estimate product recovery.
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- Conducted techno-economic analysis to assess process feasibility and industrial viability.
- Participated in HAZOP studies to identify process hazards and recommend safety improvements.
- Contributed to plant layout development considering process flow, equipment arrangement, and operational safety.
- Collaborated with faculty supervisors and technical advisors to refine process design and validate engineering assumptions.

Events Chair | KU Car Club | Khalifa University | January 2025 – May 2026

- Coordinated the planning and delivery of 5+ university automotive events, working closely with board members to manage timelines and event logistics.
- Collaborated with university departments, media teams, and club leadership on budgeting, approvals, and event execution.
- Coordinated volunteers and supported cross-functional teams to ensure smooth event operations for events with 50+ attendees.
- Built partnerships with other student organizations to organize collaborative on-campus and external events.

VOLUNTEER EXPERIENCE

Lifestyle Fair & Sustainability Event | Volunteer | September – November 2023

- Led a team of 10 students to organize interactive sustainability activities to increase student engagement and awareness of SDG initiatives engaging over 40 attendees.

ACHIEVEMENTS & CERTIFICATIONS

- 1st Place – Senior Design Project Competition, Khalifa University, 2026
- Professional IQ Program (PQ) – Mubadala, June 2025
- Dean’s List – Spring 2023, Khalifa University
- KU SDG Ambassador – 2023
- **Certificates:** Generative AI, Critical Thinking, Entrepreneurship Foundations, Portfolio Building

SKILLS

Engineering & Process Tools: Aspen HYSYS, MATLAB, Python, MS Office (Word, Excel, PPT), Canva

Engineering Skills: Process Simulation, Process Modelling, Data Analysis, Material & Energy Balances

Professional Skills: Technical Communication, Team Leadership, Analytical Thinking, Time Management

Technical Skills: Generative AI, Critical Thinking, Entrepreneurship Foundations, Portfolio Building
